

# Cover Crops

## WHY GROW THEM

Cover crops or “green manure” as they are often referred to, are an essential part of creating healthy soils in the market garden. They can play different roles and are used in different contexts, but for the most part we rely on them to protect the ground from soil erosion, improve soil structure and feed our soil web.

Cover crops are not a magic bullet for soil fertility (even if they can fix nitrogen) and will not replace other amendments like compost and natural fertilizers, but they are an important « tool » to grow abundant and beautiful crops all while sustaining fertility over time.

### 4 main functions of covers in the market garden:

- **Protection effect:** The soil needs to be covered in winter times when vegetables are not being grown. This prevents erosions of your topsoil.
- **Structuring effect:** The roots system of cover crops will dig your soil filling all the cavities and create an uncompacted soil profile.
- **Fertilizing effect:** Legumes fertilize crops with **N** without adding phosphorus to the soil.
- **Feeding effect:** The roots are feeding your soil web , in turn, fuels you systems.  
Plant to grow deep rooting crops at least once in your calendar year.

For these effects to occur, a cover crop must set on a period sufficiently long. As a general rule of thumb, your cover crops should have at least 8 to 10 weeks to develop and propagate. Another 7 to 10 days is also needed for mowed down cover crop residues to be digested before planting into.

To get the most fertilizing action out of a leguminous green manure, it helps to keep the following two pieces of information in mind:

Firstly, the best time to mix in a green manure is just before it flowers. It is at this stage that the plant has stored its maximum nitrogen and will make the greatest contribution to soil fertility. With the crop still young and tender, it will decompose quickly and easily, in time for the following crop.

Secondly, you need to know that legumes do not fix nitrogen from the air by themselves. The fixing is actually done by bacteria of the genus *Rhizobium*, which form nodules on the roots of the leguminous plants and permit the exchange to take place. However, these bacteria may not necessarily be present in the soil if legumes have not been grown there for some time. Regardless, it is a good idea to inoculate your legume seeds with the proper rhizobia.

## **COVER CROPS AND LIVING SOILS**

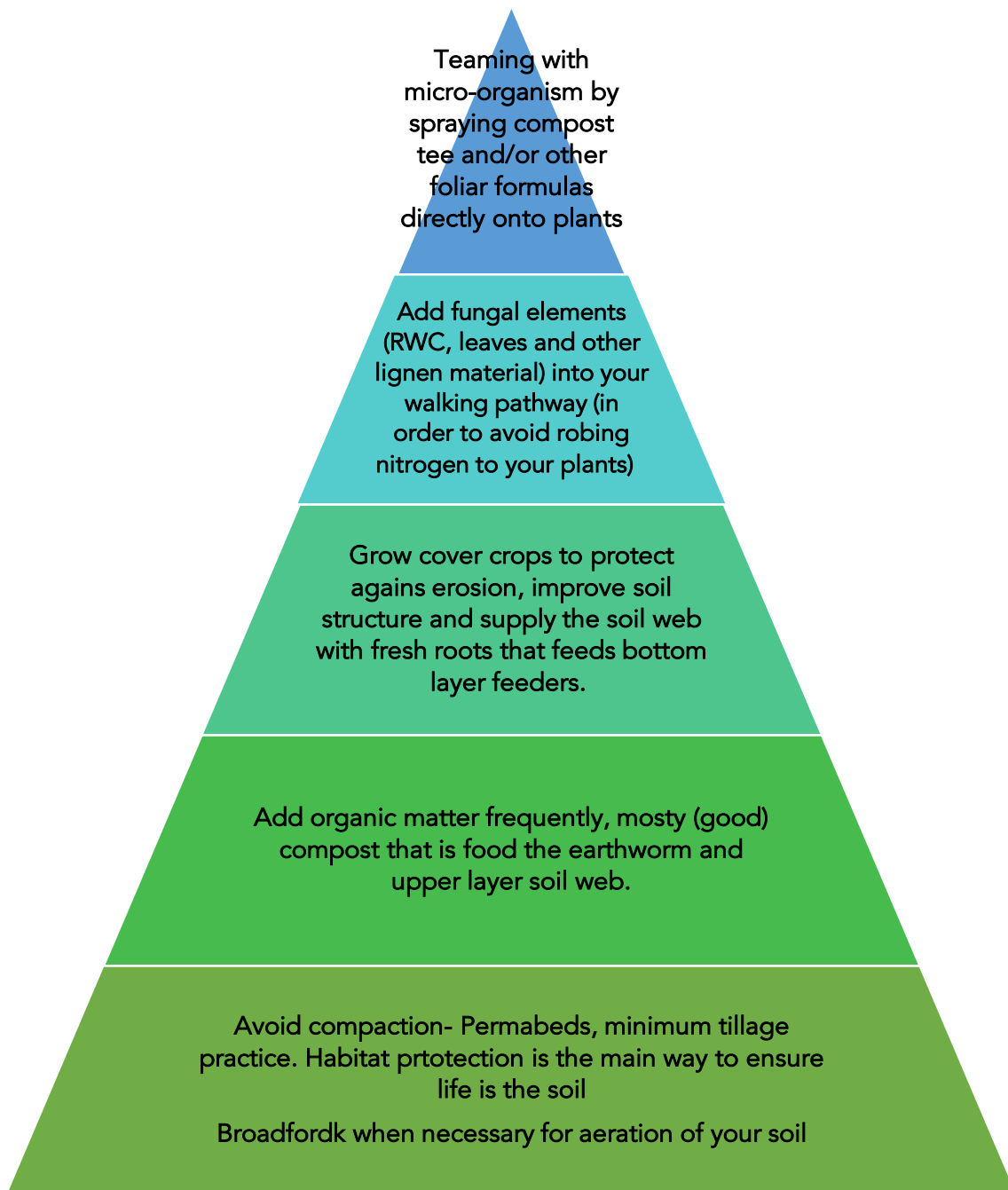
Our choice of growing cover crops and our strategies to deal with them should be understood in the context of working with living soils. Remember that the biota in the ground (ie. all the living organisms in the soil) is the main driving factor of all fertility available for your crops. The more active and alive your soil is, the more fertile it is, the better your crop will be.

Also remember that the ecosystem (ie. the habitat) in which this biota lives is fragile and in many ways influence by your farming practices. Soils that are compacted and/or continuously tilled cannot allow for earthworms (amongst others) to play a crucial role in fundamental biological processes to makes healthy soils.

Other basic principles for creating optimal life in soils all interact together and it's convenient to understand these components as hierarchical levels of a pyramid. The lowest levels conditions must be satisfied before higher up conditions can be met. The diagram on the next page illustrates the stacking role that each strategies plays in the overall fertility of your soil and can help you prioritize where to start from.

Getting back to cover crops, what's important to understand is that when we stop tilling, organic matter (the food of the soil web), doesn't get as easily brought down to the lower layers of your soil. A rototiller or plow does that very well and explains why most growers « till under » their cover crops.

# Essential Components to Build a Healthy Soil Food Web



In a minimal tillage approach, to feed your subsoil with fresh organic matter and improve nutrient cycling, we rely on earthworms, which move up and down the soil (It may be difficult to believe, but every particle of soil in healthy soils has passed through the intestines of earthworms ! ) AND on decaying roots of deep rooted cover crops that have populated the bottom layer of your top soil. Cover crops improve not only the physical aspect of your soil, but the soil web dynamics. Get it? good!

## **WHICH ONE TO USE ?**

There is a myriad of different plant species that can be used as cover crops and all kind of context how and why they are used. A reliable source of information for selecting the best cover crop for your farm is the SARE book on cover crops. ([Managing Cover Crops Profitably, 3rd Edition, available online for free](#))

Another great resource is [The Cover Crop Seed Comparison Chart](#) developed by Johnny's Selected Seeds. This chart allows you to sort different seeds according to hardiness zones also providing you with minimum germinating temperatures.

In all cases, I suggest mixing different species of legumes and cereals and spreading at a rate of least ¼ bucket per 100' (30M) bed. The application rate is much higher than what's written on labels, and the intention here is to cover the bed surface as fast as possible. We do this to limit the proliferation of weeds on the bed.

## **HOW TO GROW THEM**

### **Implantation Steps:**

- Prepare bed by using tarps or in any case, start with beds as "clean" as possible
- Broadfork the bed, going as deep as possible
- In a bucket or wheel barrel, mix different seeds using preferred ratio
- To kickstart the growth we add to the mix a powdered fertilizer like crab or alfalfa meal. When seeding legumes on new ground, it might be a good idea to inoculate your seeds with the proper rhizobium\*\*

- Spread seeds by broadcasting by hand. A drilling device such an Earthway seeder or fertilizer spreader can also be used, by we deem it not to be necessary.
- Bury seeds using the rototiller, lifting it upward to only cultivate to top inches (+/- 5 cm) of the surface. This step is necessary for the seeds, not the dry up. Alternatively, you can scratch the surface of your bed with your bed prep rake.
- Install irrigation and moisten the beds thoroughly.
- If possible put Ramial Wood Chips or leafs in the walkways of your permabeds. This will add a fungal element to your soil.
- When ready destroy the cover using strategy described below

*\*Inoculants are living organism that you can order from seed companies. They come is the from of a black powder and with an expiration date. Treat the inoculant as you might treat a living organism—don't leave it in the sun for extended periods of time, and store it in a cool dry place when not in use, such as a refrigerator. Follow the inoculant manufacturer's directions on the package for proper mixing rate.*

## **TURNING THEM UNDER**

The end game is to destroy the cover crop without disturbing the ecology that has now been created by the cover crop.

- This is done using a flail mower that chops the the plant material into a mulch that will be easily digested by the soil web of your bed.
- Then using the rotary plow, bury the bed with at least 0,5" (1cm) of soil.
- Cover the bed with silage tarp and let rest for +/- 2 weeks, depending on how warm the soil is. (in spring and fall, when soils are cold this process takes more time).
- The objective is to let the microorganism in your soil diggest the buried cover crop before planting into it.
- Resume regular implantation procedures.

## HOW TO WORK COMPACTED SOILS ?

Compacted soil can be fixed by using both the power of a tractor and the deep penetrating effect of deep rooting cover crop. The overall strategy is to try and open the soil below the hardpan (mechanically using a chisel plow or ripper or with a longtime broafork) You open the soil as deep as possible in order to allow the roots of the cover crop to establish themselves as deep as possible. In this scenario, cover crops should be kept in the ground for at least 10 weeks or as long as possible.

Adding lots of organic of compost and mixing it in the first 10" (25cm) of your soil using a rototiller can also be helpful to help colonize your soil with both organic matter and soil with micro-organisms.

### Steps:

- Work from a clean slate. Beds have been tarped or any case no perennial weeds are established.
- Open up the soil using a heavy duty broadfork or Vibro\*. At that point, you want to go as deep as possible to break the hardpan.
- Add compost or manure, a rough estimate of at least 2"(5cm) on the surface.
- Using the rototiller, mix the compost into the soil, doing it in 2 passes. The goal is to bring the compost (manure) in the bottom layer of the bed.
- Seed your cover crops broadcasting the seed at the determine application rate. The seeds can be inoculated with mycorrhizal inoculant previously.
- Bury seeds using the rototiller, lifting it upward to only cultivate to top inches (+/- 5 cm) of the surface. This step is necessary for the seeds, not the dry up. Alternatively, you can scratch the surface of your bed with your bed prep rake.
- Install irrigation and moisten the beds thoroughly.
- The fastest the roots reach the compacted zone, the better. The longer the cover crop stays in the ground the better.

- When ready destroy the cover using strategy described above.

*\*If the compaction problem is severe and a hardpan can be found as deep as 12" (30cm) deep, the broadfork will need to be replaced by a ripper that is tractor mounted. If the wheels of the tractor are not spaced around 48" (120cm) center to center (ie the tractor rolls in the pathways, overriding the bed), The permanent beds will need to be rebuilt when the cover crops is turned over. You don't make an omelet without breaking eggs.*